

Management of brain metastases in a patient with HER2-positive GEJ cancer and long-term survival

ICI/AFROC Joint Virtual Monthly Tumor Board
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Arturo Lecumberri, MD, MSc
Hospital Universitario de Navarra
Navarrabiomed-IdiSNA

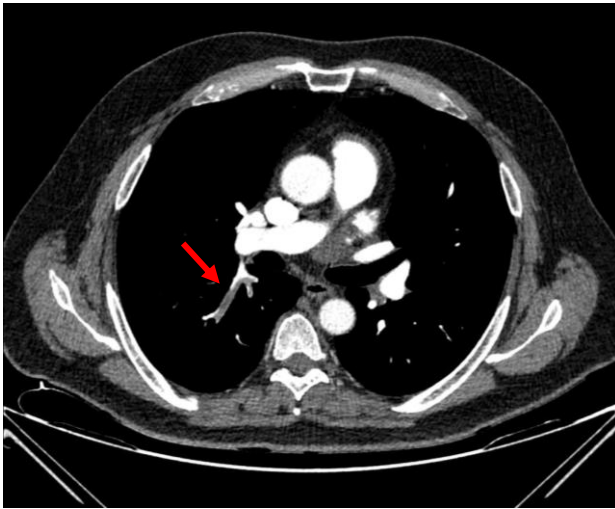


59 year old male.

Previous medical history of colonic diverticulosis, appendectomy and cholecistectomy.

No alcohol or tobacco consumption.

Consults at the ER after 2 weeks of progressive dyspnea.



Low risk, bilateral pulmonary embolism.



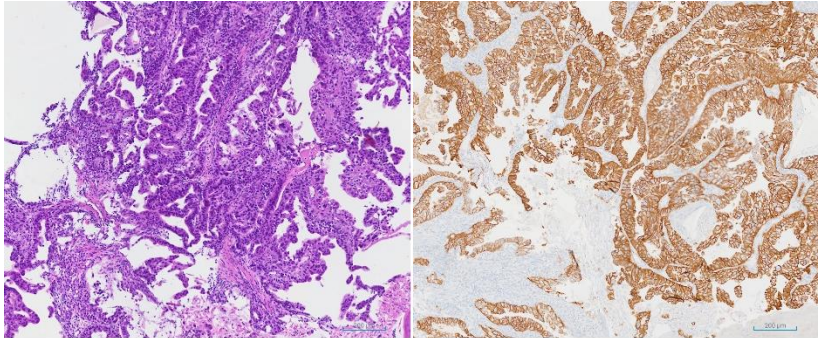
Blood tests performed during admission show iron deficiency anemia.

Diagnostic workup for anemia is started.



Gastroscopy

Irregular, friable lesion with a 15-20 mm diameter length.



Pathology report

Moderately differentiated, tubular adenocarcinoma. Conserved expression of mismatch repair proteins (MLH1, MSH2, MSH6 and PMS2). HER2 IHC 3+.



CT scan

Thickening of the distal oesophagus with extension to stomach.
Massive metastatic liver spread. Large tumoral masses in both liver lobules.
Extense node dissemination (supra- and infradiaphragmatic adenopathies).

59 year old previously healthy male diagnosed with **MMRp, HER2-positive, stage IV** (liver, non-locoregional adenopathies) **GEJ** tubular **adenocarcinoma**

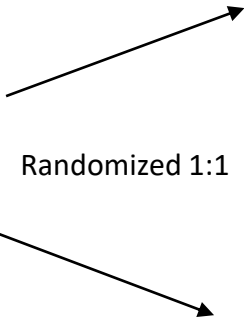
1st line
systemic therapy

ToGA

Open-label, multicentre, international, phase 3, randomised controlled trial.

Stratified by ECOG PS, ChT regimen, extent of disease, primary cancer site, and measurability of disease.

Adult patients with HER2+ (IHC 3+ or FISH positive), inoperable LA/recurrent/metastatic gastric or GEJ adenocarcinoma with no previous ChT for metastatic disease
(n = 584)



Trastuzumab 6 mg/kg Q3W (loading dose 8 mg/kg on C1)
+ ChT (FP or XP) Q3W up to 6 cycles
(n = 294)

ChT (FP or XP) Q3W up to 6 cycles
(n = 290)

	Surgical specimen staining pattern	Biopsy specimen staining pattern	HER2 overexpression assessment
0	No reactivity or membranous reactivity in <10% of tumour cells	No reactivity or no membranous reactivity in any tumour cell	Negative
1+	Faint or barely perceptible membranous reactivity in ≥10% of tumour cells; cells are reactive only in part of their membrane	Tumour cell cluster with a faint or barely perceptible membranous reactivity irrespective of percentage of tumour cells stained	Negative
2+	Weak to moderate complete, basolateral or lateral membranous reactivity in ≥10% of tumour cells	Tumour cell cluster with a weak to moderate complete, basolateral or lateral membranous reactivity irrespective of percentage of tumour cells stained	Equivocal
3+	Strong complete, basolateral or lateral membranous reactivity in ≥10% of tumour cells	Tumour cell cluster with a strong complete, basolateral or lateral membranous reactivity irrespective of percentage of tumour cells stained	Positive

HER2=human epidermal growth factor receptor 2 (also known as ERBB2).

Table 1: Immunohistochemistry scoring for HER2 in gastric and gastro-oesophageal junction cancer, by type of diagnostic specimen

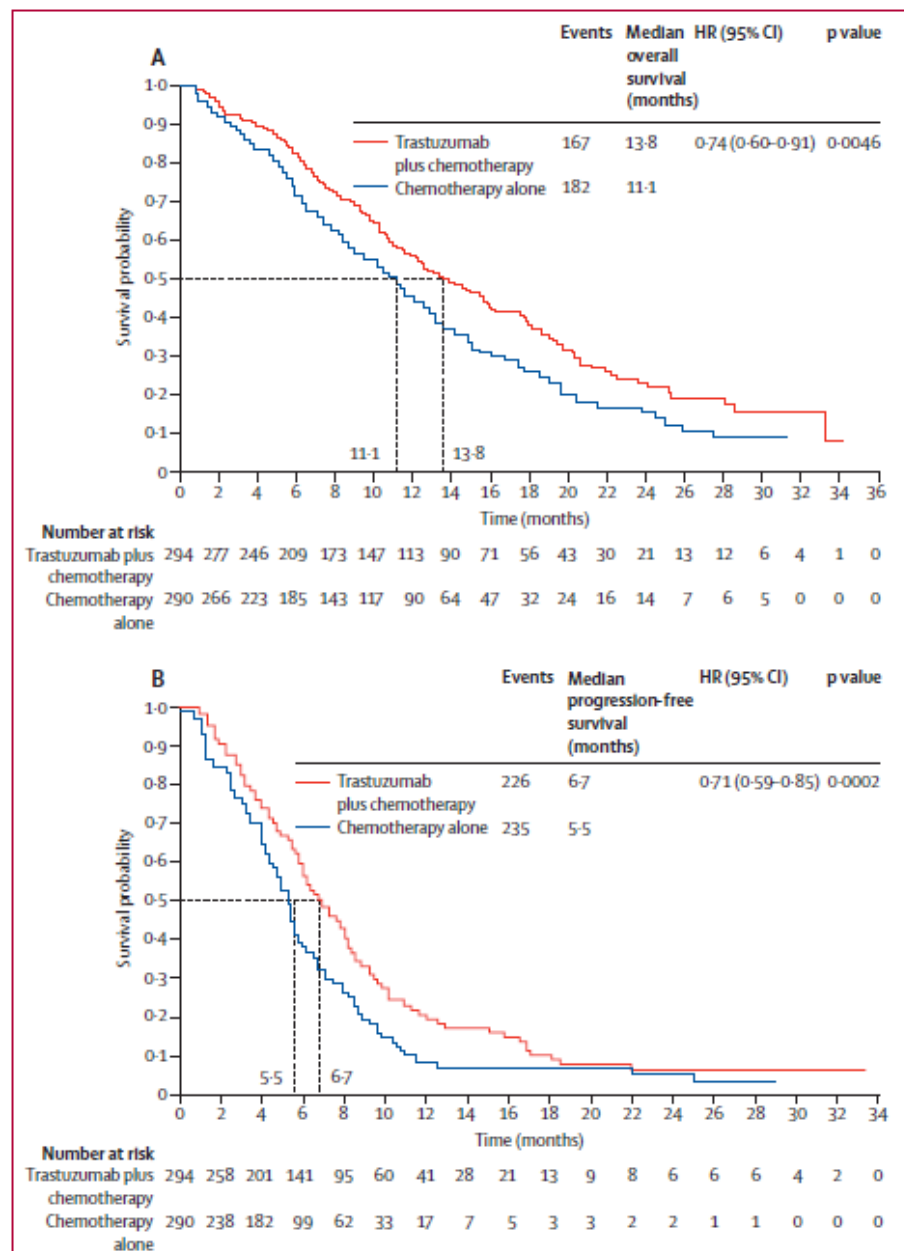


Figure 2: (A) Median overall survival and (B) progression-free survival in the primary analysis population
HR=hazard ratio.

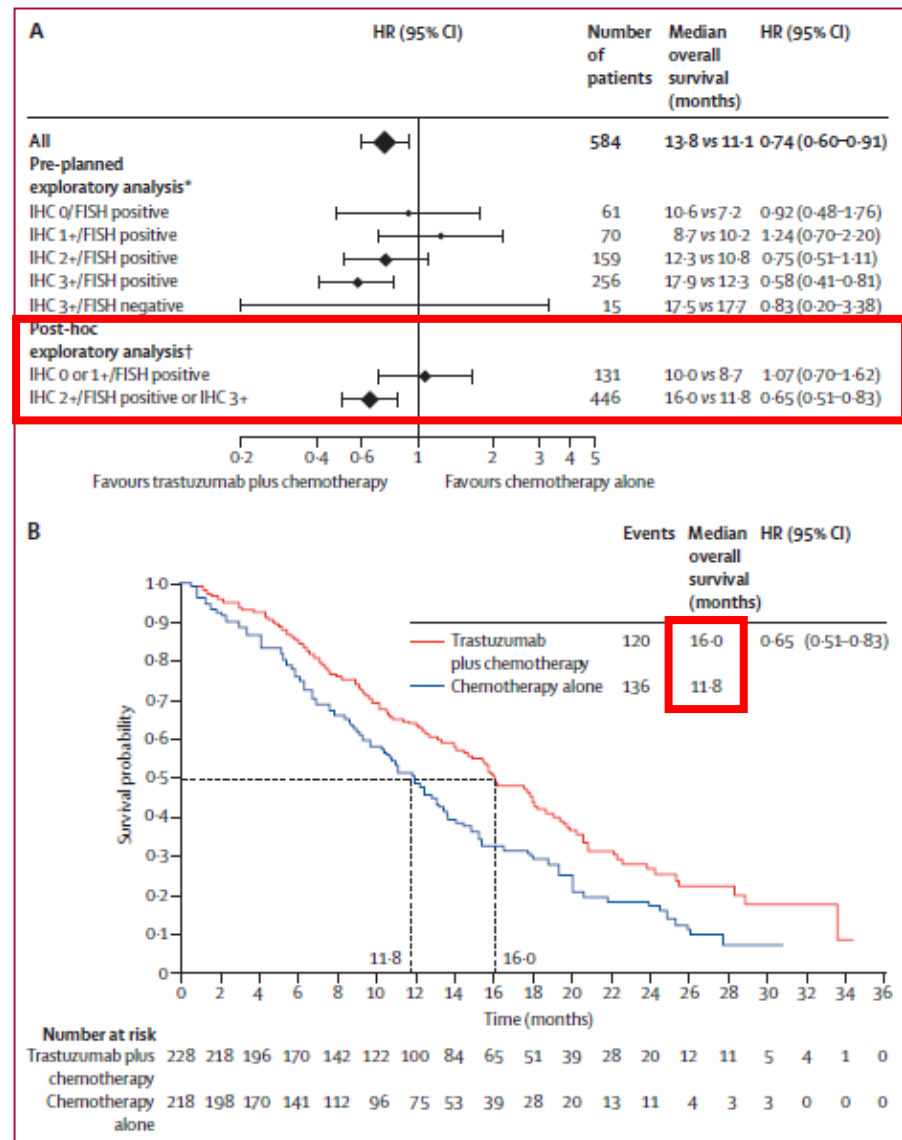


Figure 4: Exploratory analyses

HR=hazard ratio. (A) Pre-planned exploratory and post-hoc exploratory analyses of patients stratified by human epidermal growth factor receptor 2 (HER2) status. *n=561; patients with no immunohistochemistry (IHC) data (n=7) or IHC 3+ tumours with no fluorescence in-situ hybridisation (FISH) data (n=16) were excluded from this analysis. †n=577; patients with no IHC data (n=7) were excluded from this analysis. (B) Overall survival according to the post-hoc exploratory analysis (FISH and IHC) in patients with IHC 2+ and FISH-positive tumours or IHC 3+ tumours.

KEYNOTE-811

Randomised, multicenter, international, double-blind, placebo-controlled, phase 3 trial.

Stratified by geographical region, PD-L1 CPS (≥ 1 vs < 1) and ChT regimen.

Adult patients with previously untreated unresectable or metastatic, pathologically confirmed HER2-positive (IHC 3+ or IHC 2+ with positive ISH) gastric or GEJ adenocarcinoma
(n = 698)

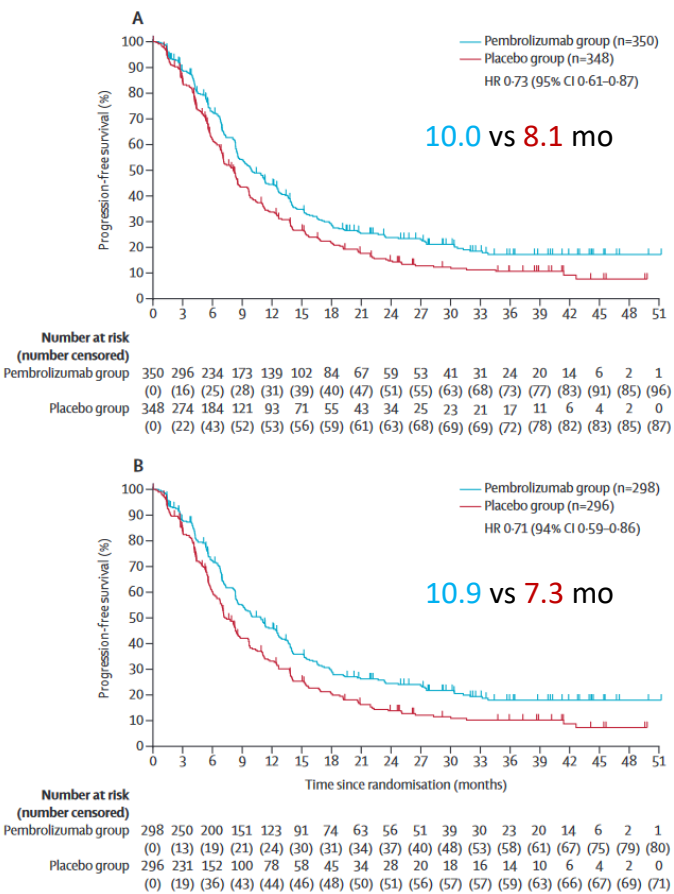
Randomized 1:1

Pembrolizumab 200 mg IV Q3W +
Trastuzumab 6 mg/kg IV Q3W (loading dose 8 mg/kg on C1)
+ ChT (FP or CAPOX) Q3W
(n = 350)

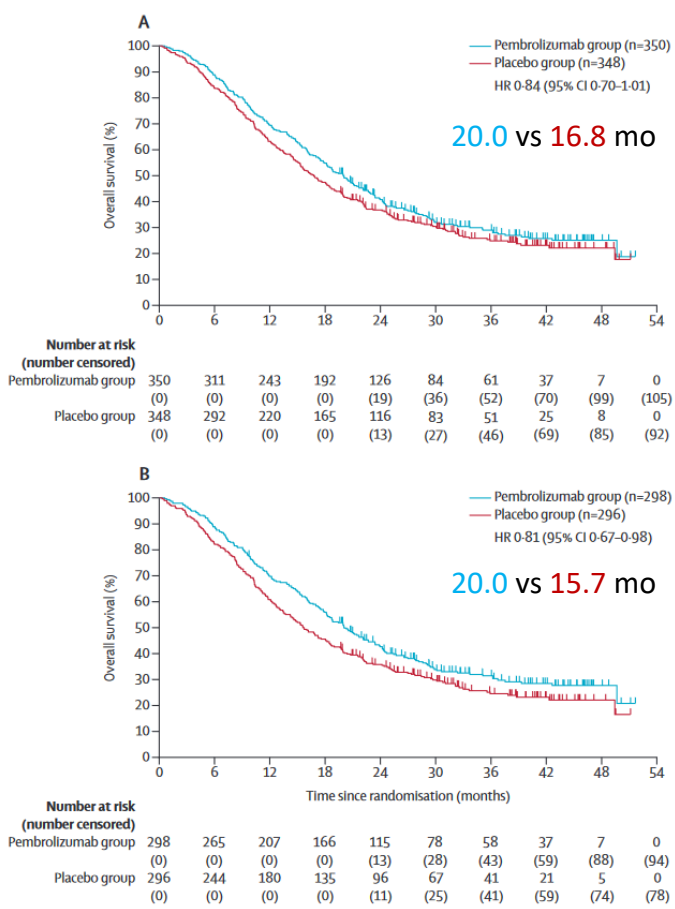
Placebo IV Q3W +
Trastuzumab 6 mg/kg IV Q3W (loading dose 8 mg/kg on C1)
+ ChT (FP or CAPOX) Q3W
(n = 348)

	Pembrolizumab group (N=350)		Placebo group (N=346)	
	Any	Grade ≥ 3	Any	Grade ≥ 3
Any adverse event	347 (99%)	248 (71%)	346 (100%)	225 (65%)
Any treatment-related adverse event*	341 (97%)	204 (58%)	334 (97%)	176 (51%)
Serious	88 (25%)	76 (22%)	79 (23%)	66 (19%)
Led to death	4 (1%)	4 (1%)	3 (1%)	3 (1%)
Led to discontinuation of any drug	124 (35%)	59 (17%)	108 (31%)	44 (13%)
Any adverse event of interest†	132 (38%)	36 (10%)	83 (24%)	12 (3%)

KEYNOTE-811

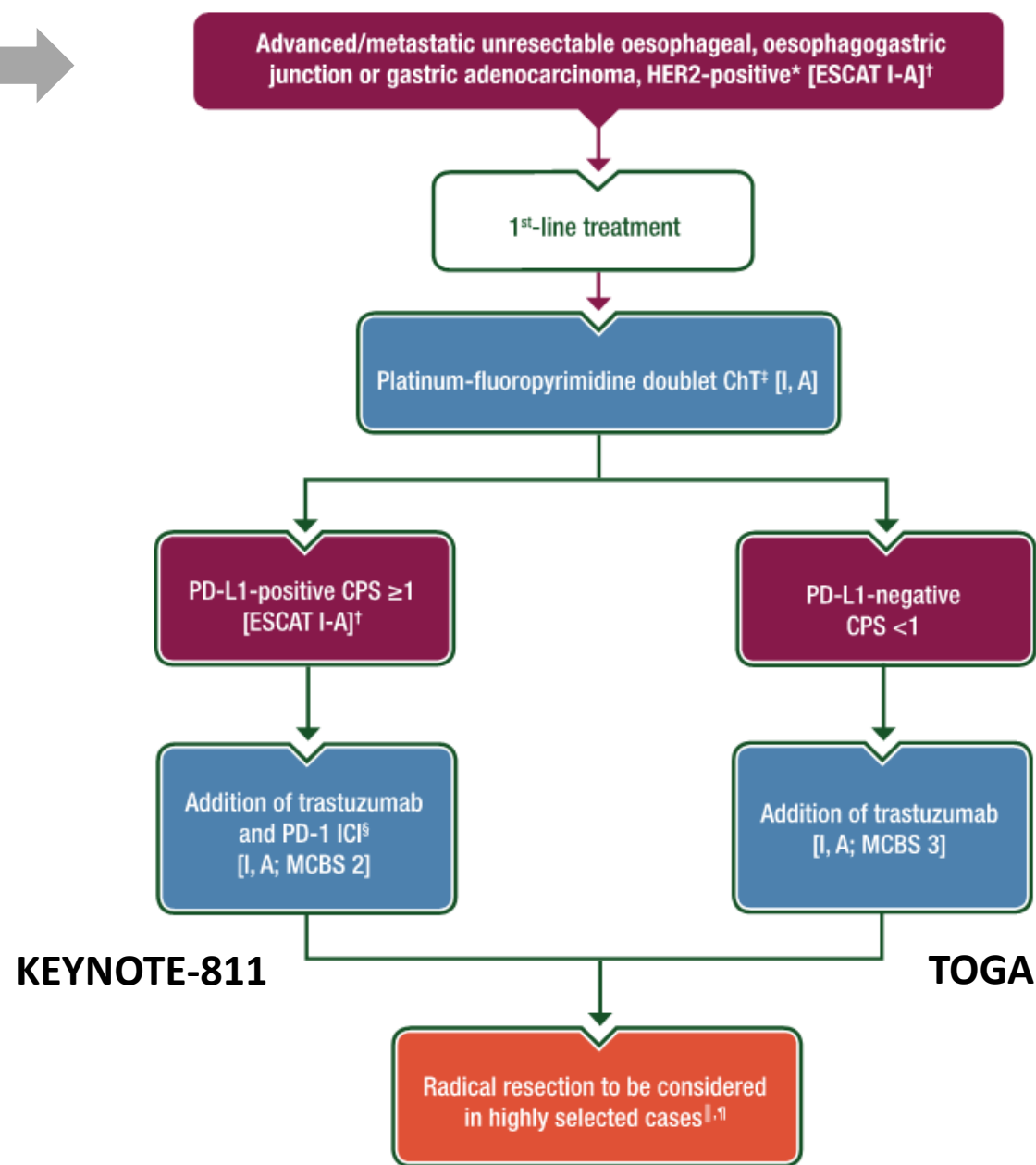
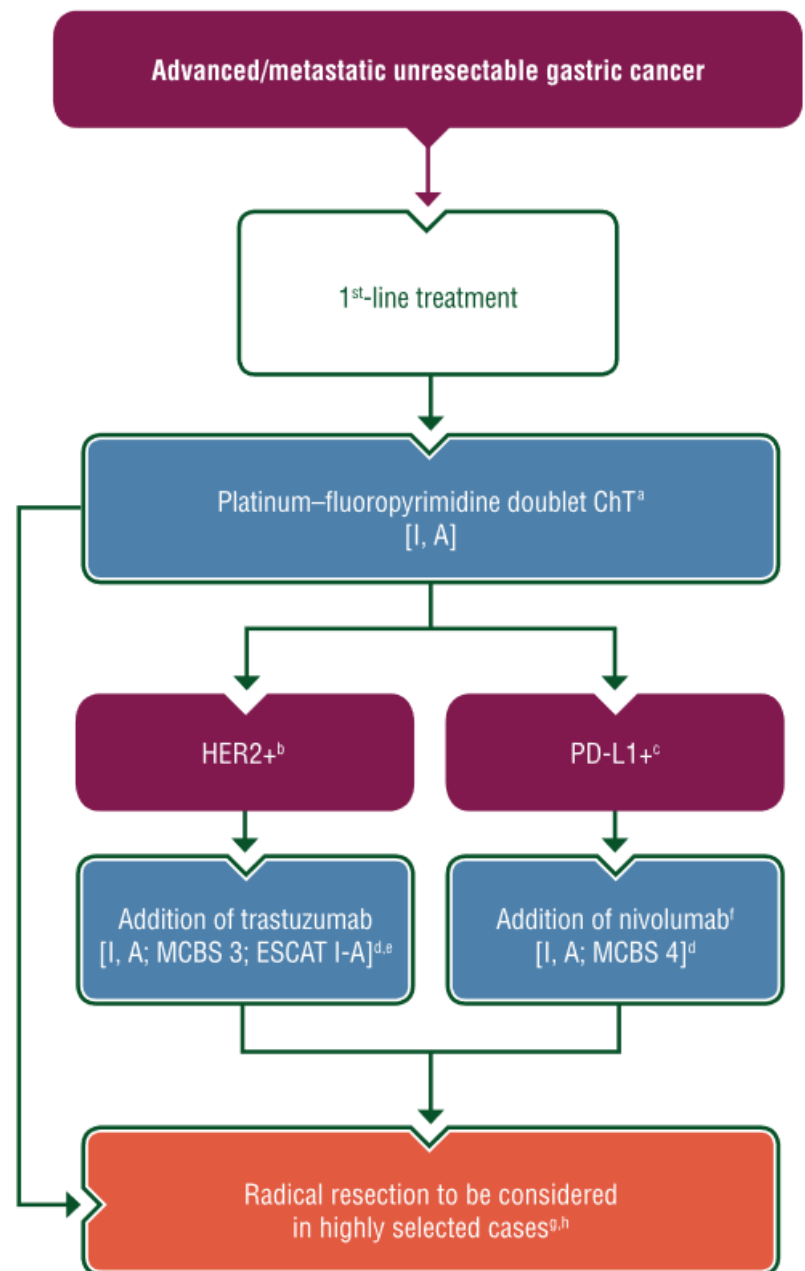


PFS in ITT population.
A) All patients.
B) Patients with CPS≥1 tumors.

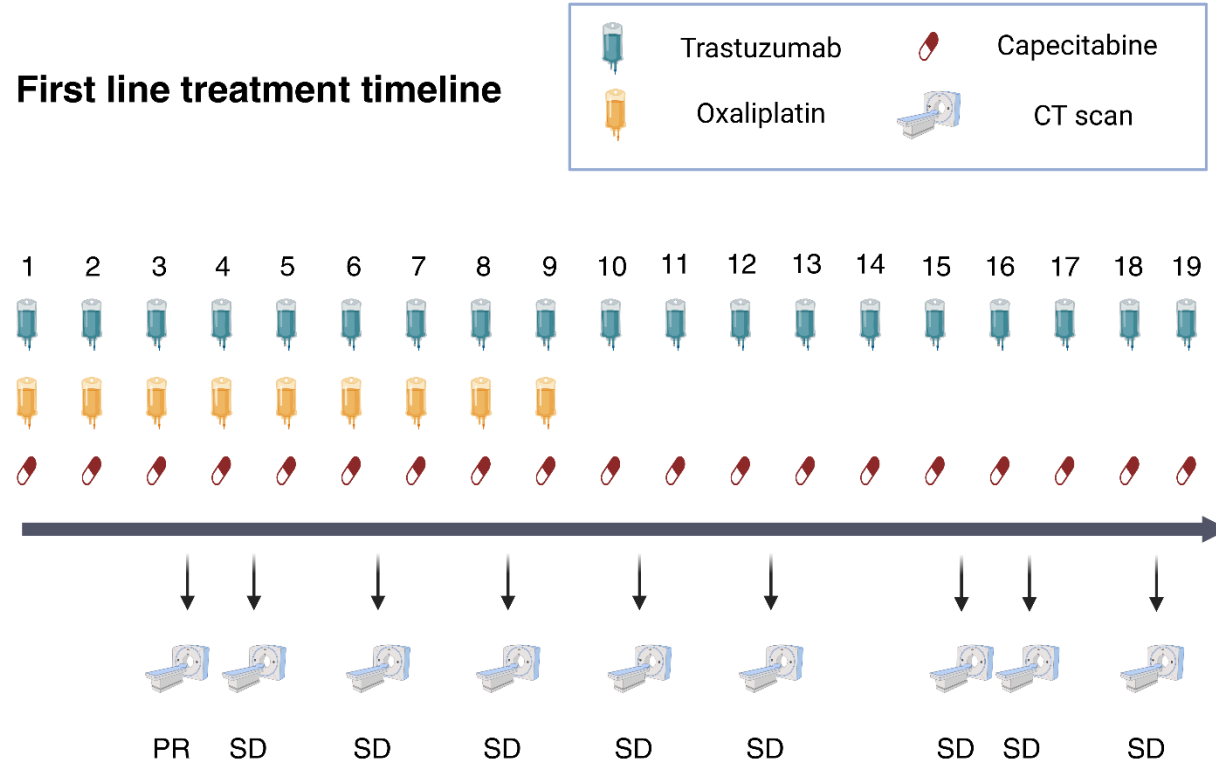


OS in ITT population.
A) All patients.
B) Patients with CPS≥1 tumors.

	Events/patients, n/N		HR (95% CI)
	Pembrolizumab group	Placebo group	
Age, years			
<65	152/205	153/192	0.67 (0.54-0.85)
≥65	101/145	108/156	0.84 (0.64-1.10)
Sex			
Female	42/66	55/68	0.49 (0.32-0.74)
Male	211/284	206/280	0.83 (0.69-1.01)
Race			
Asian	76/119	80/121	0.85 (0.62-1.16)
Non-Asian	177/231	179/225	0.69 (0.56-0.84)
Geographical region			
Europe, North America, and Australia	84/113	88/111	0.73 (0.54-0.99)
Asia	75/118	78/119	0.84 (0.61-1.16)
Rest of world	94/119	95/118	0.65 (0.49-0.87)
PD-L1 status			
CPS≥1	217/298	225/296	0.71 (0.59-0.86)
CPS<1	36/52	36/52	1.03 (0.65-1.64)
MSI status			
Non-high	234/326	244/329	0.75 (0.63-0.90)
ECOG performance status			
0	101/146	101/146	0.76 (0.58-1.00)
1	152/204	160/202	0.74 (0.59-0.92)
Primary tumour location			
Stomach	171/240	173/226	0.70 (0.57-0.87)
Gastro-oesophageal junction	82/110	88/122	0.85 (0.63-1.15)
Histological subtypes			
Diffuse	50/68	44/51	0.71 (0.47-1.07)
Intestinal	136/198	134/188	0.73 (0.57-0.92)
Indeterminate	66/83	83/109	0.81 (0.59-1.12)
Tumour size (above median)			
Yes	133/171	130/170	0.71 (0.56-0.91)
No	108/162	125/166	0.74 (0.57-0.96)
Number of metastatic sites			
≤2	123/179	146/198	0.71 (0.56-0.91)
≥3	130/171	115/150	0.78 (0.61-1.00)
Previous gastrectomy			
Yes	33/51	45/63	0.81 (0.52-1.27)
No	220/299	216/285	0.72 (0.59-0.87)
Chemotherapy regimen			
CAPOX	211/297	222/299	0.74 (0.61-0.90)
FP	42/53	39/49	0.73 (0.47-1.14)
Overall	253/350	261/348	0.73 (0.61-0.87)



First line treatment timeline



Best response: partial response (60% reduction in target lesions)

Tumor marker normalization: from 196 ng/mL (prior to 1st line start) to 2 ng/mL.

Relevant toxicity: G2 peripheral sensory neuropathy (rel. OXL), G1-2 gastrointestinal toxicity (rel. capecitabine).

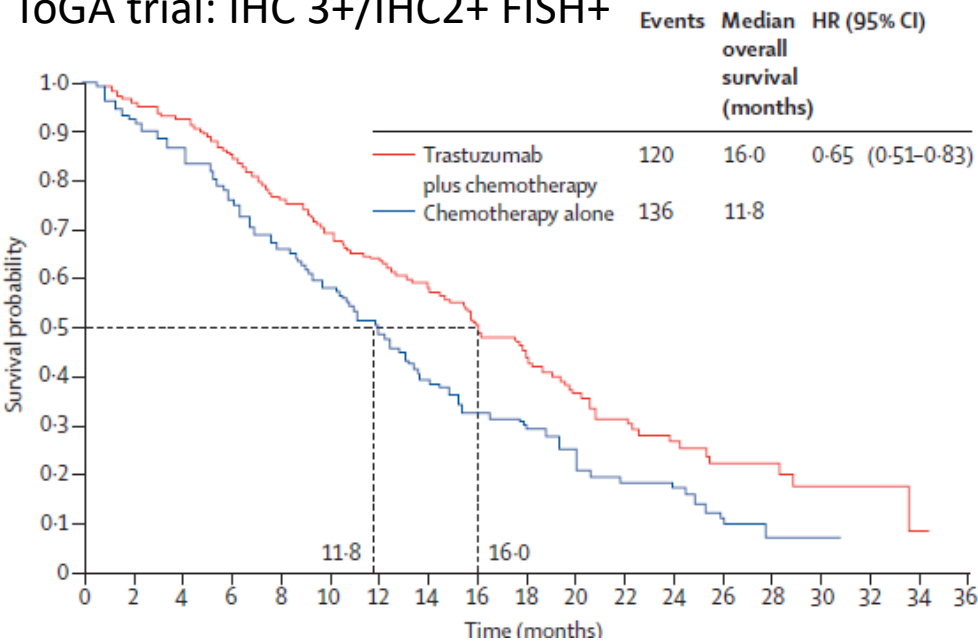


VARIANZ study

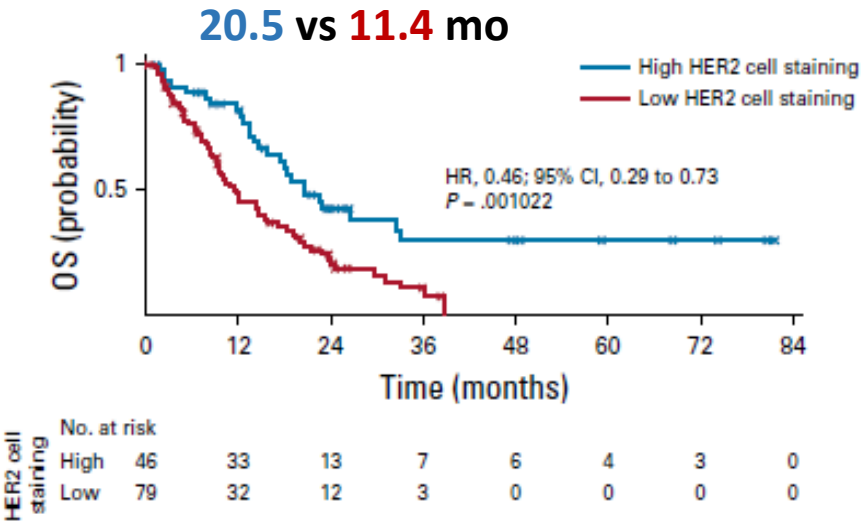
Patients receiving medical treatment for histologically confirmed mGC (including EGJC) were recruited. HER2 status was determined centrally. Patients were grouped according to local + central HER2 status.

To suggest alternative criteria for HER2-directed treatment, thresholds optimizing the statistical significance for survival benefit were determined.

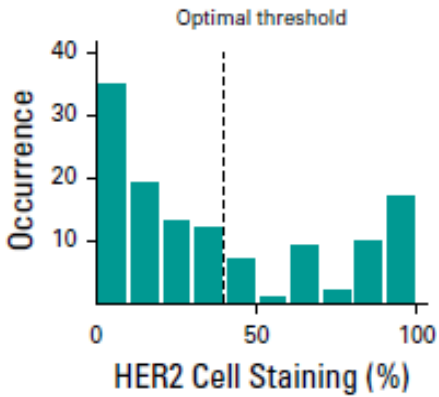
ToGA trial: IHC 3+/IHC2+ FISH+



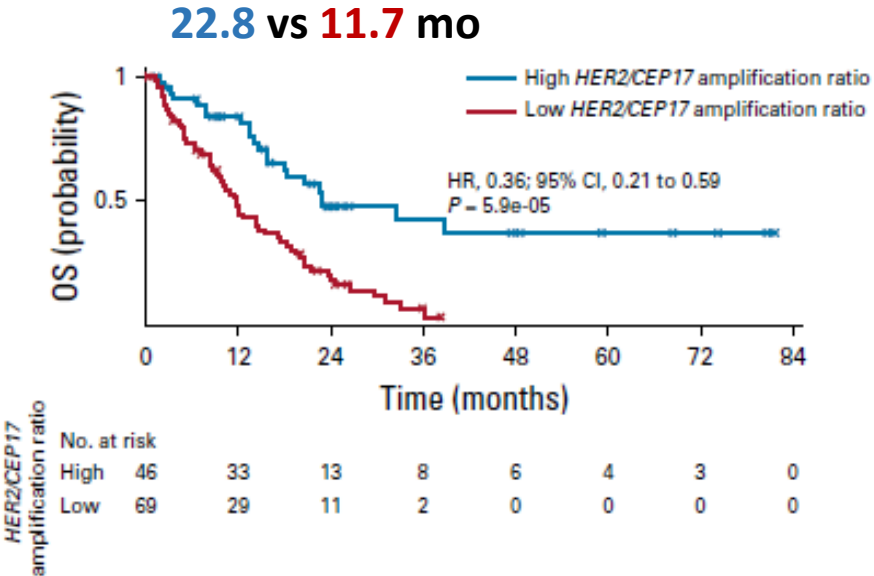
A



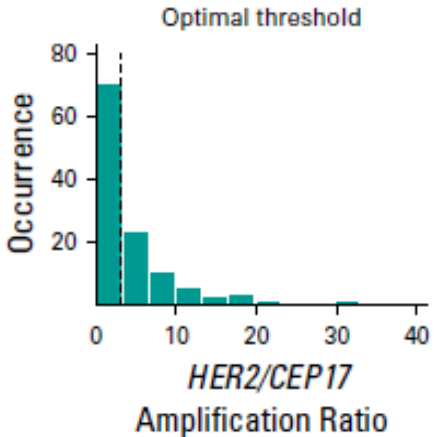
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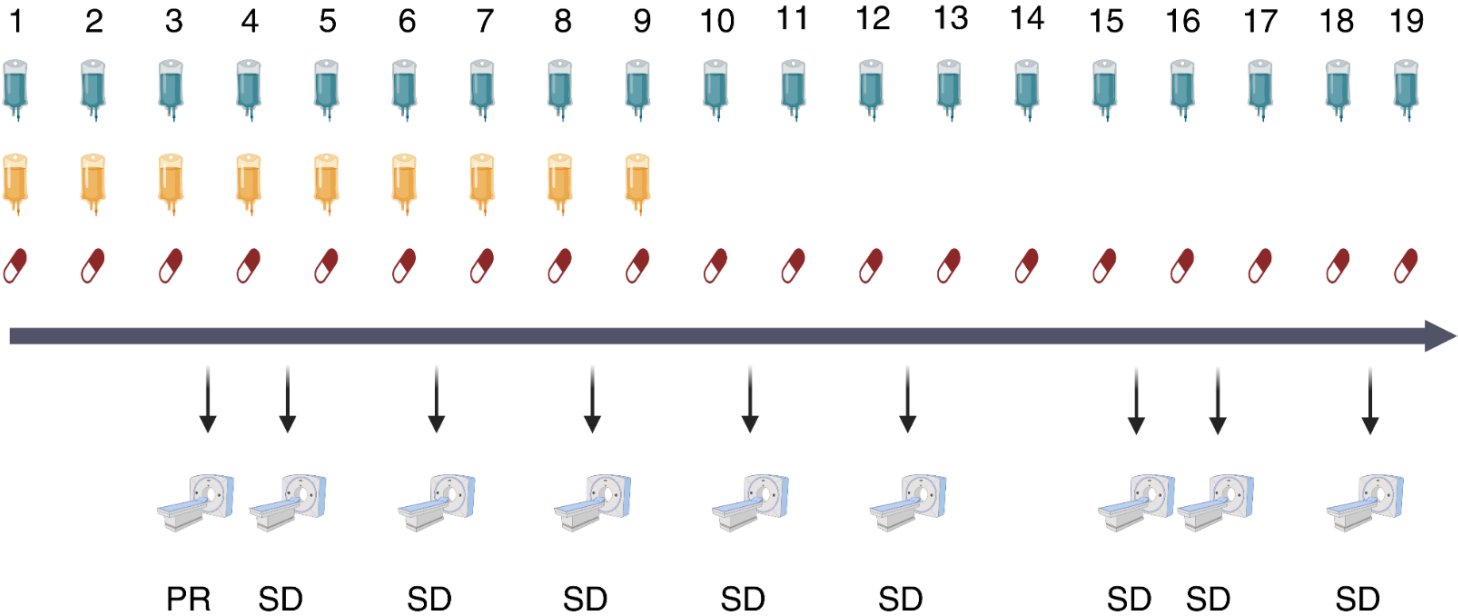
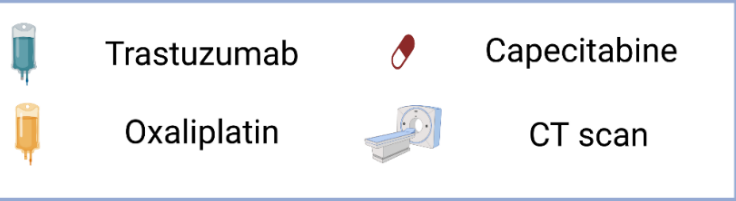
C



D

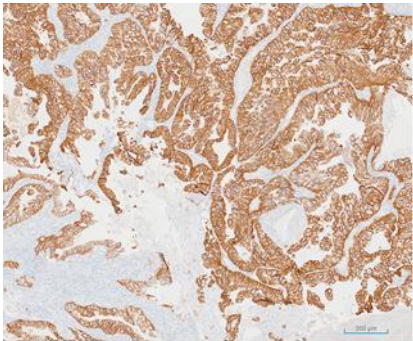


First line treatment timeline



Consults at the ER after
cephalea, dizziness,
photophobia and vomiting

Disease progression (BMs)
PFS 13,3 mo

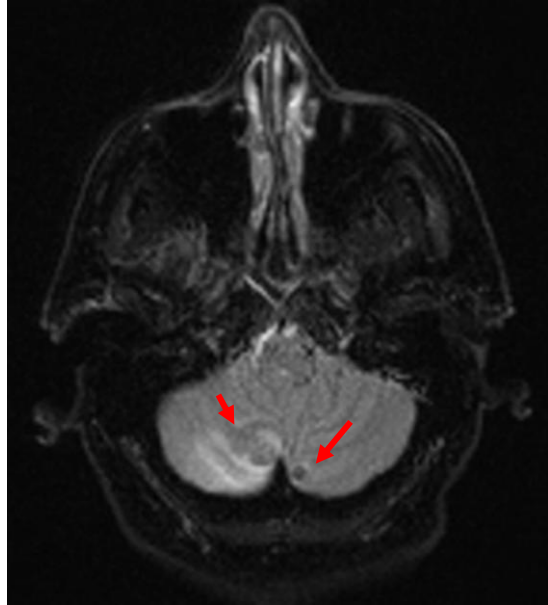


HER2 expression
in almost 100% of
tumor cells



Brain CT scan

Compatible with **hemorrhagic metastasis** in the right cerebellar hemisphere. Signs of **ascending transtentorial herniation** are observed.

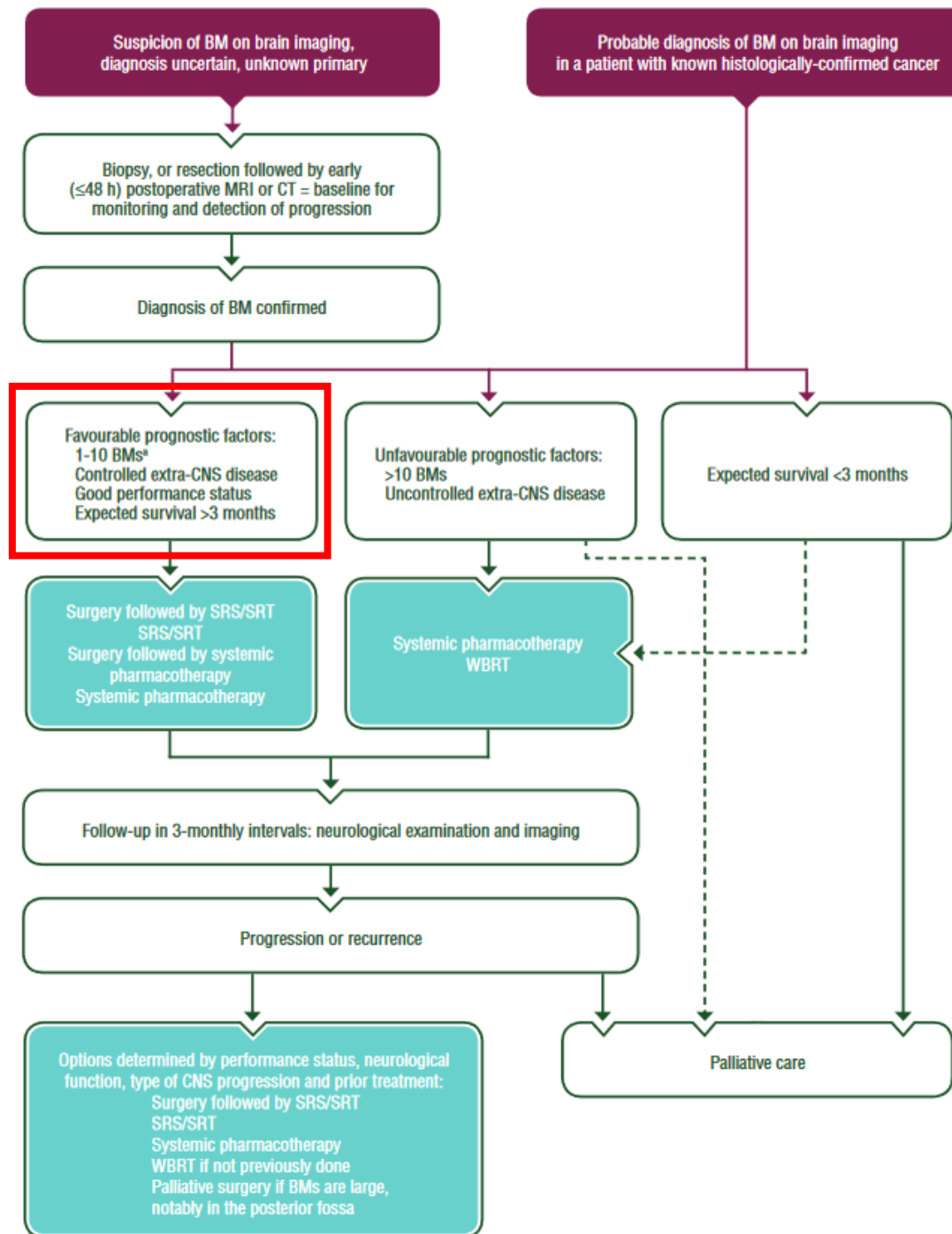


Brain MRI

Paramedian right cerebellar lesion (28x26 mm) with cystic areas and hemorrhagic component. The lesion produces **mass effect** and slight **IV ventricle shift**.

An **additional lesion (7x8,4 mm)** with similar characteristics is identified in the **posterioinferior** region of the **left cerebellum**.

Both lesions are compatible with a diagnosis of **cerebellar metastasis**.



Multidisciplinary Tumor Board

Surgical treatment for the cerebellar lesions is proposed.

Both cerebellar lesions were **surgically removed** (R0; biopsy compatible with gastric cancer metastasis).

The patient underwent **post-operative radiotherapy (25 Gy)** targeting the **surgical bed**.

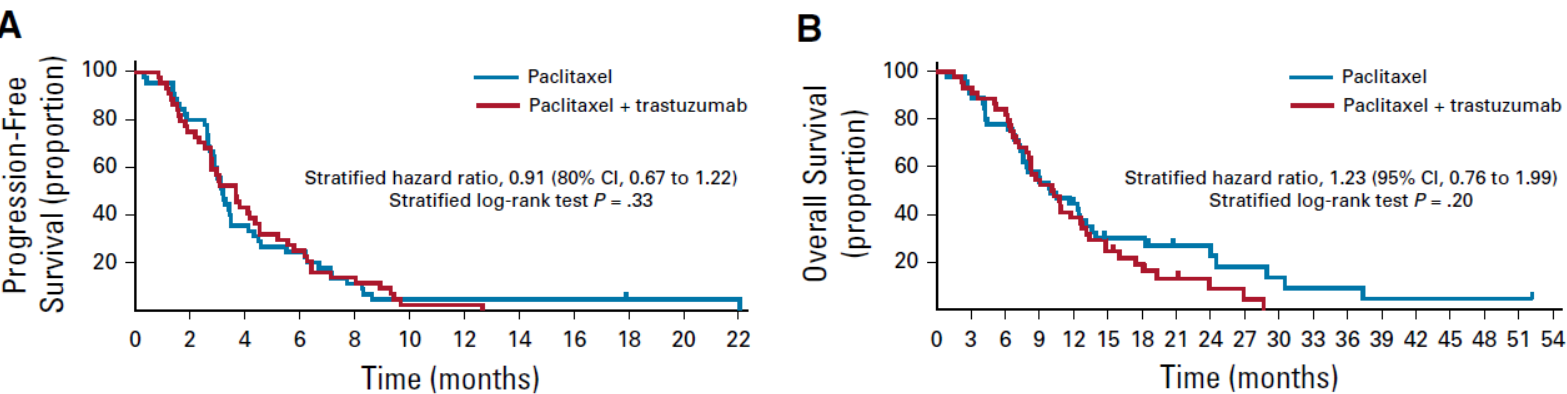
**2nd line
systemic therapy**

2nd line
systemic therapy

T-ACT Study

Open-label, multicenter, randomized phase II trial.
Stratified by center, ECOG PS, HER2 status and presence of measurable lesions.

Evaluates the continuation of trastuzumab (paclitaxel vs paclitaxel + trastuzumab) in patients with HER2 positive G/GEJ tumors that had progressed to prior treatment with platinum, fluoropyrimidine and trastuzumab.



No. at risk:													No. at risk:																			
Paclitaxel	45	36	16	11	5	2	2	2	2	1	1	1	Paclitaxel	45	41	35	25	19	10	10	6	6	4	3	2	2	1	1	1	1	1	0
Paclitaxel + trastuzumab	44	33	19	11	6	1	1	0	0	0	0	0	Paclitaxel + trastuzumab	44	41	37	24	17	10	7	4	2	1	0	0	0	0	0	0	0	0	0

****NEED FOR RE-BIOPSY!!**

	Paclitaxel arm (n = 10)									Paclitaxel+trastuzumab arm (n = 8)								
IHC before first-line therapy																		
FISH before first-line therapy																		
Tumor response for first-line therapy	PR	PR	PR	PR	PR	PR	PR	PR		PR	PR	PR	PR	PR	PR	PR		
IHC after first-line therapy																		
FISH after first-line therapy																		
Circulating HER2 DNA amplification																		
Tumor response					PR		PR			PR					PR			

3+ for IHC or positive for FISH or circulating HER2 DNA amplification

2+ for IHC

1+ for IHC

0 for IHC or negative for FISH or circulating HER2 DNA amplification

Not evaluated

Not evaluable

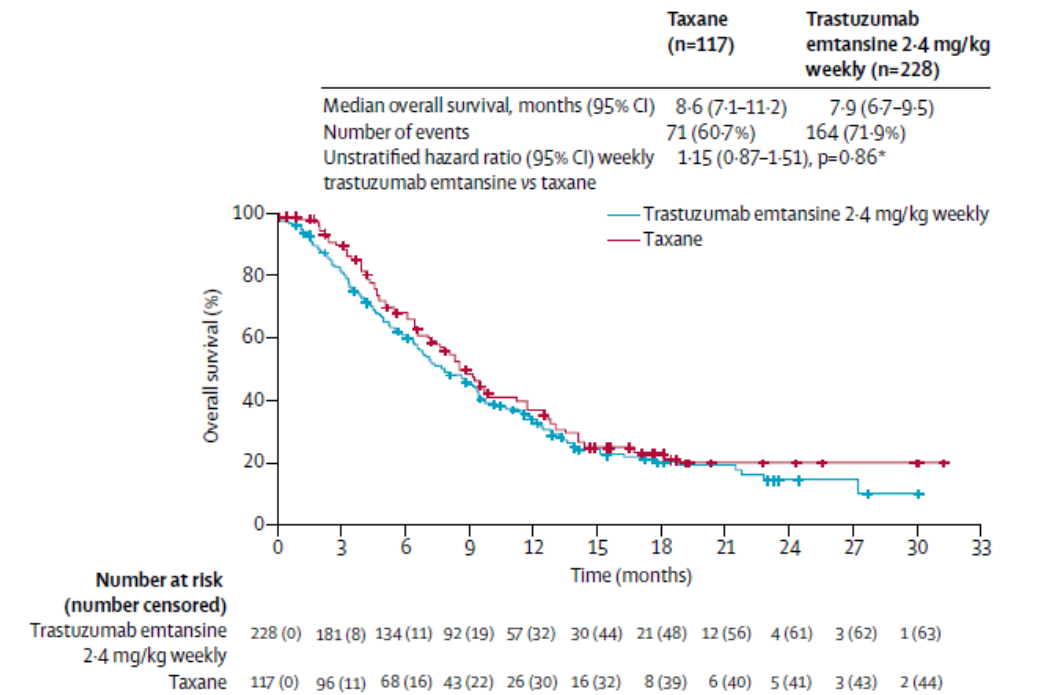
PR

Stable disease or non-CR/non-PD

PD

GATSBY trial

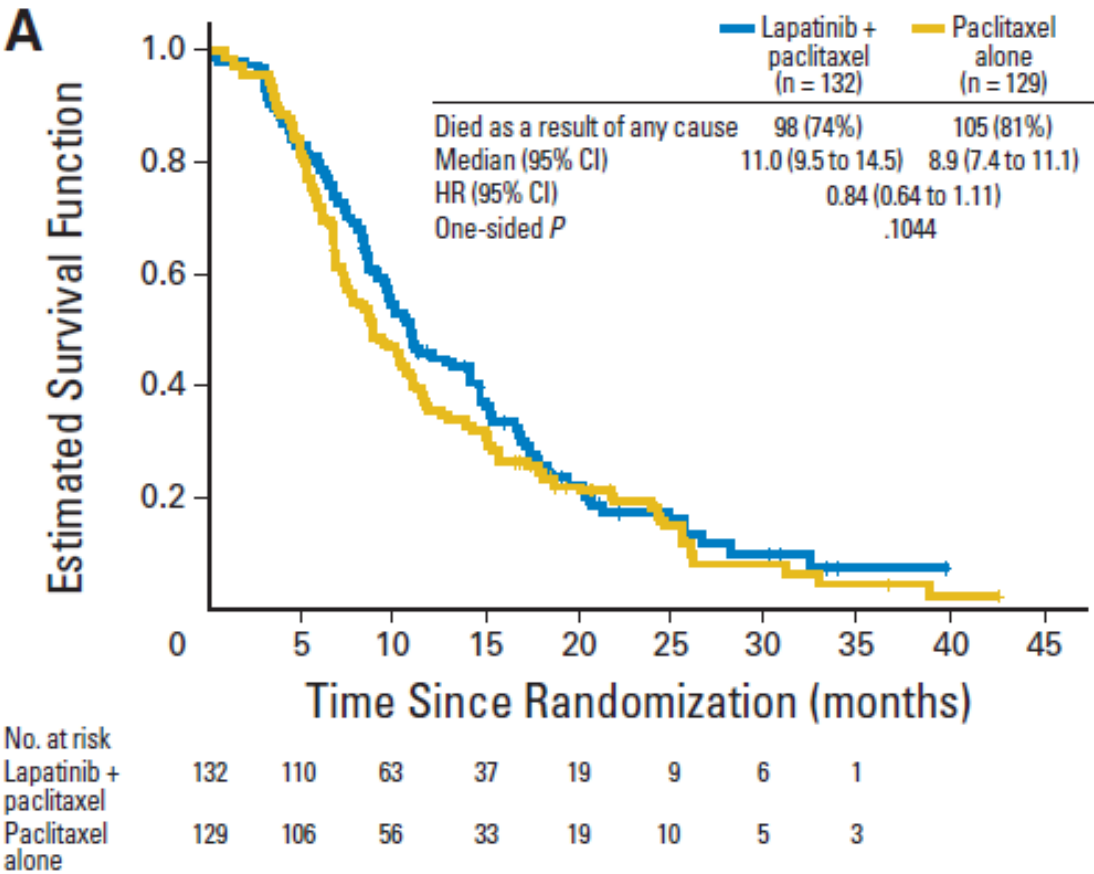
Randomised, open-label, adaptive, phase 2/3 study.
Eligible patients had HER2-positive advanced gastric cancer and progressed during or after first-line therapy. Compares 2nd-line T-DM1 vs physician’s choice of a taxane.



	Taxane (n=102)	Trastuzumab emtansine 2-4 mg/kg weekly (n=204)	Difference	p value (χ²)
Objective response	20 (19.6%)	42 (20.6%)	0.98% (-9.04 to 11.00)	0.8406 (two-sided)
Median duration of response* (months)	3.7 (2.8-5.6)	4.3 (3.0-6.8)	..	..

TyTAN trial

Phase 3 study: dose optimization phase and randomized phase.
Eligible patients had HER2-positive (FISH HER2/CEP17 ratio ≥2) gastric cancer that had progressed to prior therapy. Patients received lapatinib + paclitaxel vs paclitaxel alone.



DESTINY-Gastric01

Open-label, multicenter, randomized, phase 2 trial.
Stratified by country (Japan or South Korea), ECOG PS
and HER2 status.

Adult patients with HER2+ LA/metastatic
gastric or GEJ adenocarcinoma that had
progressed to ≥2 previous regimens
N = 188

Randomized 2:1

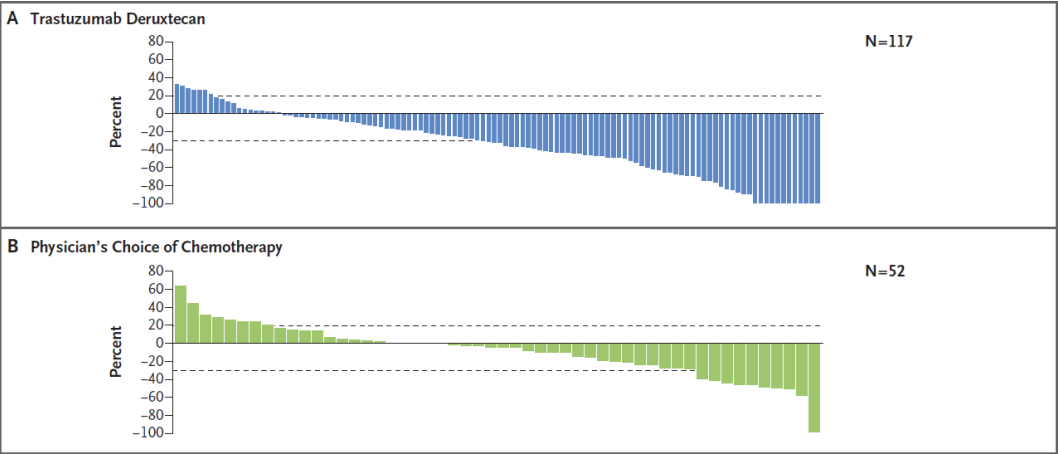
T-Dxd 6,4 mg/kg Q3W
(n = 125)

Irinotecan 150 mg/m2 Q2W or
Paclitaxel 80 mg/m2 D1, 8, 15 Q4W
(n = 62)

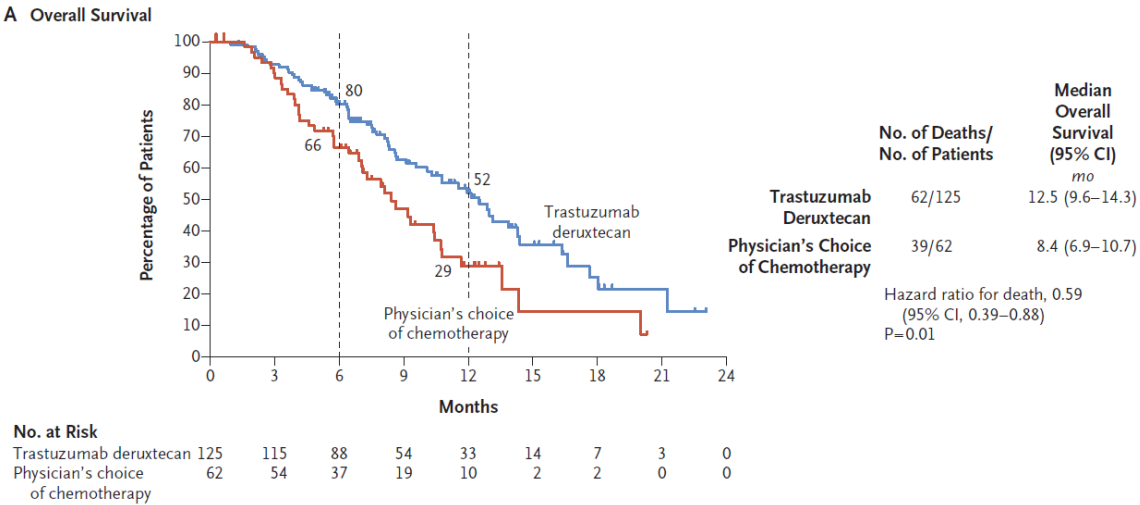
Table 3. Adverse Events Occurring in at Least 20% of the Patients Treated with Trastuzumab Deruxtecan.*

Preferred Term	Trastuzumab Deruxtecan (N=125)			Physician's Choice of Chemotherapy (N=62)		
	Any Grade	Grade 3	Grade 4	Any Grade	Grade 3	Grade 4
number of patients (percent)						
Nausea	79 (63)	6 (5)	0	29 (47)	1 (2)	0
Neutrophil count decreased†	79 (63)	48 (38)	16 (13)	22 (35)	10 (16)	5 (8)
Decreased appetite	75 (60)	21 (17)	0	28 (45)	8 (13)	0
Anemia‡	72 (58)	47 (38)	0	19 (31)	13 (21)	1 (2)
Platelet count decreased§	49 (39)	12 (10)	2 (2)	4 (6)	1 (2)	1 (2)
White-cell count decreased¶	47 (38)	26 (21)	0	22 (35)	5 (8)	2 (3)
Malaise	43 (34)	1 (1)	0	10 (16)	0	0
Diarrhea	40 (32)	3 (2)	0	20 (32)	1 (2)	0
Vomiting	33 (26)	0	0	5 (8)	0	0
Constipation	30 (24)	0	0	14 (23)	0	0
Pyrexia	30 (24)	0	0	10 (16)	0	0
Alopecia	28 (22)	0	0	9 (15)	0	0
Fatigue	27 (22)	9 (7)	0	15 (24)	2 (3)	0
Lymphocyte count decreased	27 (22)	8 (6)	6 (5)	2 (3)	0	1 (2)

Interstitial lung disease in 10% of the
patients: most cases were G1-2 (9/12),
2 were grade 3, and 1 was grade 4.



ORR: 51% (T-Dxd) vs 14% (ChT)

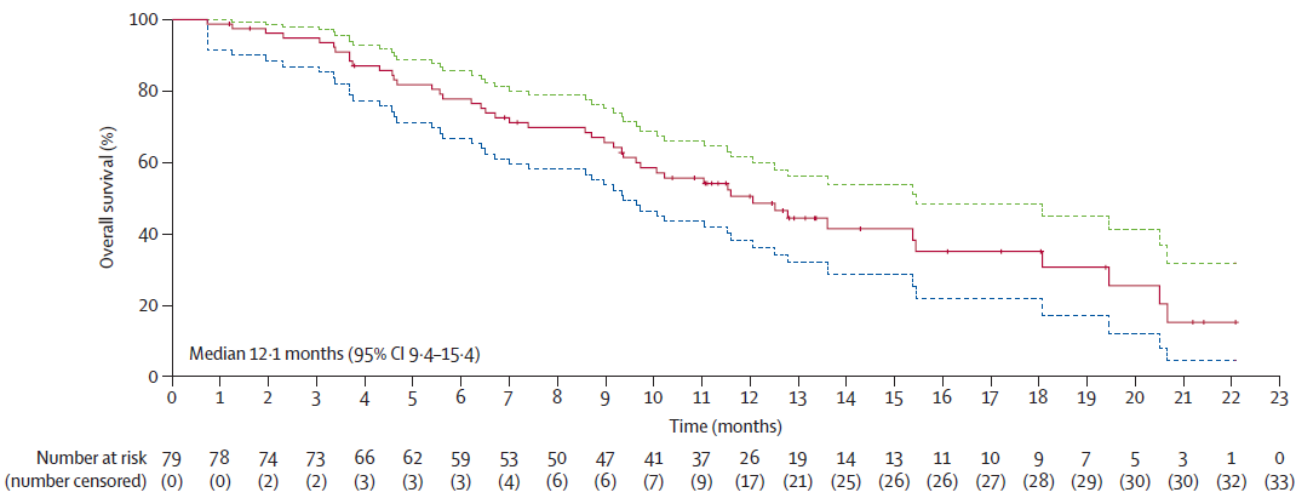
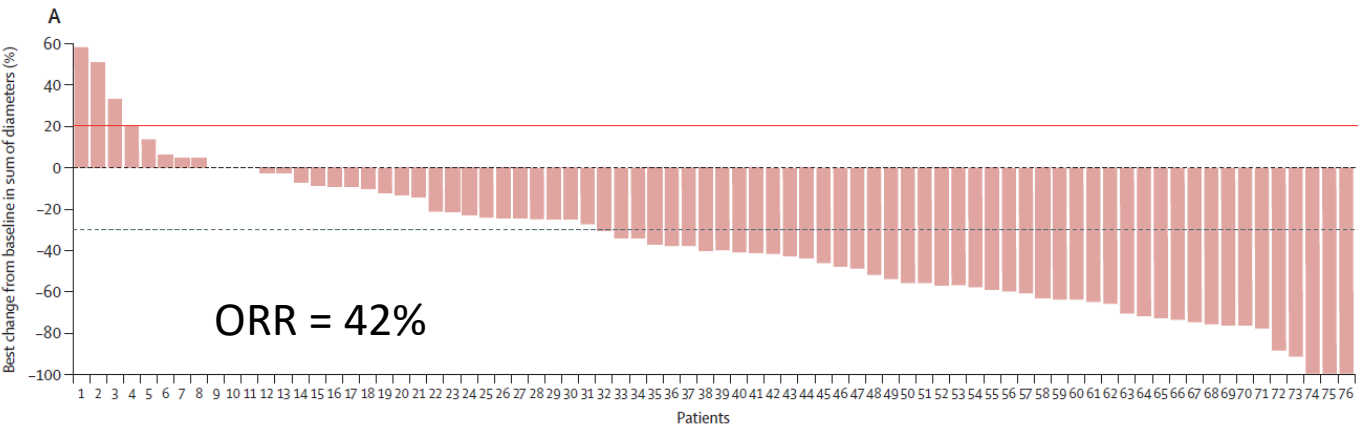


DESTINY-Gastric02

Multicentre, single-arm, phase 2 study.

Adult patients with HER2+ unresectable/metastatic gastric or GEJ adenocarcinoma that had progressed on or after first-line therapy with a trastuzumab-containing regimen (n = 79)

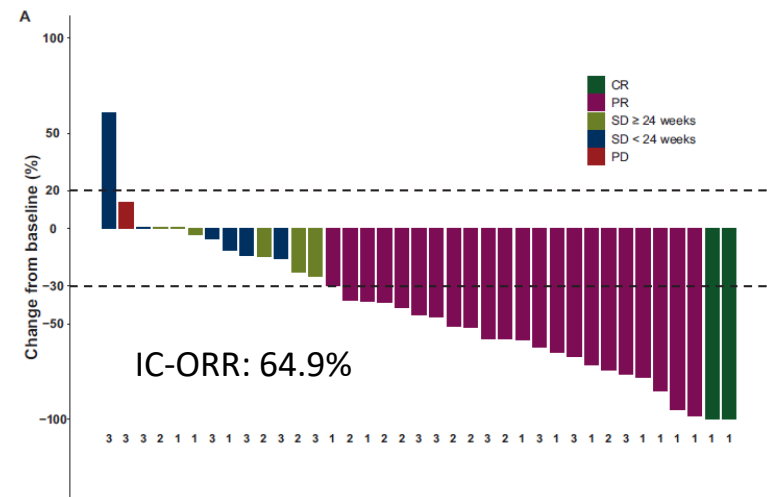
T-Dxd 6,4 mg/kg Q3W



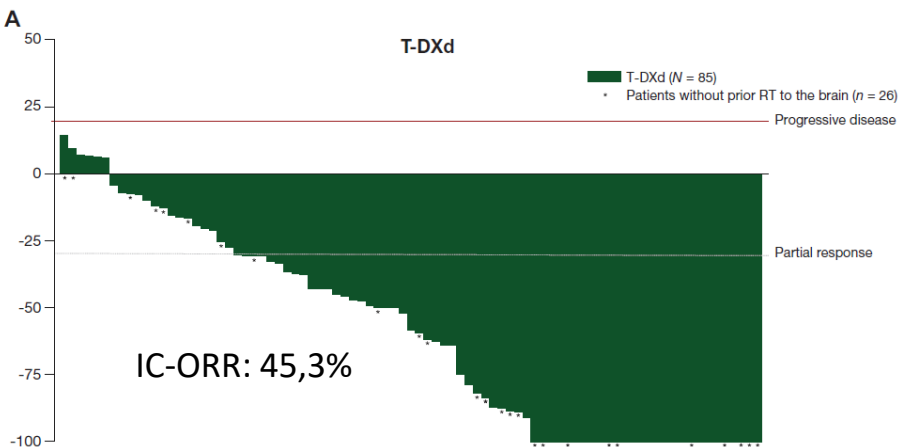
	Grade 1-2	Grade 3	Grade 4	Grade 5
Any treatment-emergent adverse event	35 (44%)	28 (35%)	5 (6%)	11 (14%)
Nausea	47 (59%)	6 (8%)	0	0
Vomiting	33 (42%)	2 (3%)	0	0
Fatigue	30 (38%)	3 (4%)	0	0
Diarrhoea	28 (35%)	1 (1%)	0	0
Decreased weight	25 (32%)	3 (4%)	0	0
Constipation	23 (29%)	0	0	0
Decreased appetite	22 (28%)	4 (5%)	0	0
Anaemia	19 (24%)	11 (14%)	0	0
Alopecia	19 (24%)	0	0	0
Decreased platelet count	12 (15%)	2 (3%)	0	0
Increased aspartate aminotransferase	12 (15%)	1 (1%)	0	0
Hypokalaemia	12 (15%)	0	1 (1%)	0
Abdominal pain	11 (14%)	2 (3%)	0	0
Asthenia	11 (14%)	1 (1%)	0	0
Pyrexia	9 (11%)	0	0	0
Blood alkaline phosphatase increased	8 (10%)	1 (1%)	0	0
Cough	8 (10%)	1 (1%)	0	0
Epistaxis	8 (10%)	0	0	0
Gastro-oesophageal reflux disease	8 (10%)	0	0	0
Hypoalbuminaemia	8 (10%)	0	0	0
Pneumonitis	7 (9%)	0	0	1 (1%)
Decreased neutrophil count	7 (9%)	2 (3%)	4 (5%)	0
Increased alanine aminotransferase	7 (9%)	1 (1%)	0	0

T-Dxd activity in breast cancer BM

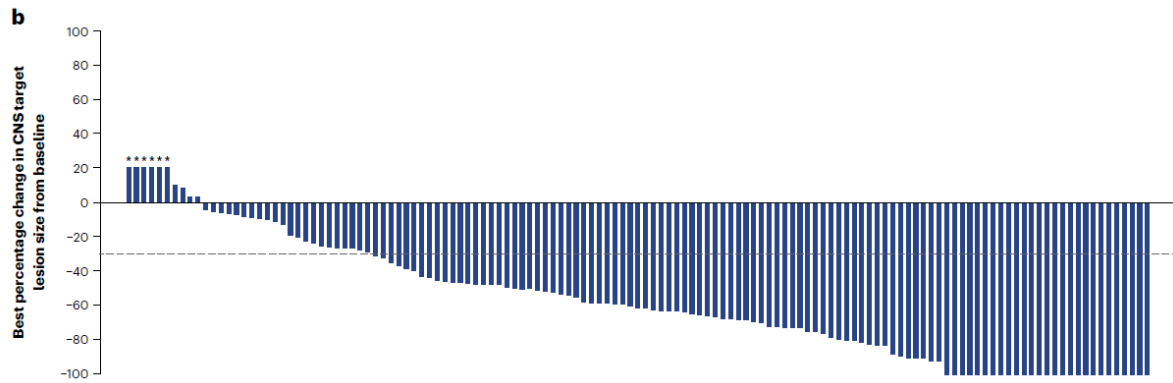
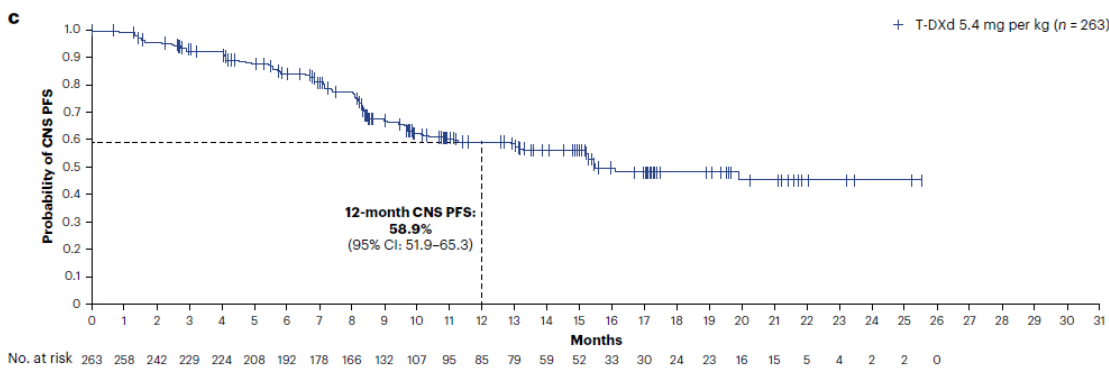
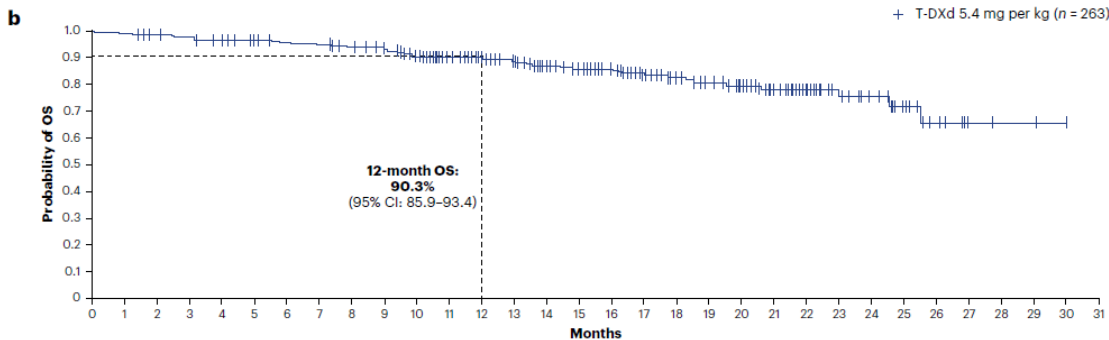
Patient-level pooled analysis of the prospective phase II DEBBRAH and TUXEDO-1 trials and the retrospective DFCI/Duke/MDACC cohort.



Pooled analysis from DESTINY-Breast01/-02/-03



Phase 3b/4 DESTINY-Breast12: T-DXd in patients with HER2+ mBC



Following C1 the patient is admitted due to G3 fatigue, hematological toxicity (G3 anemia, G4 neutropenia, G1 thrombopenia) and febril neutropenia.

Second line treatment timeline



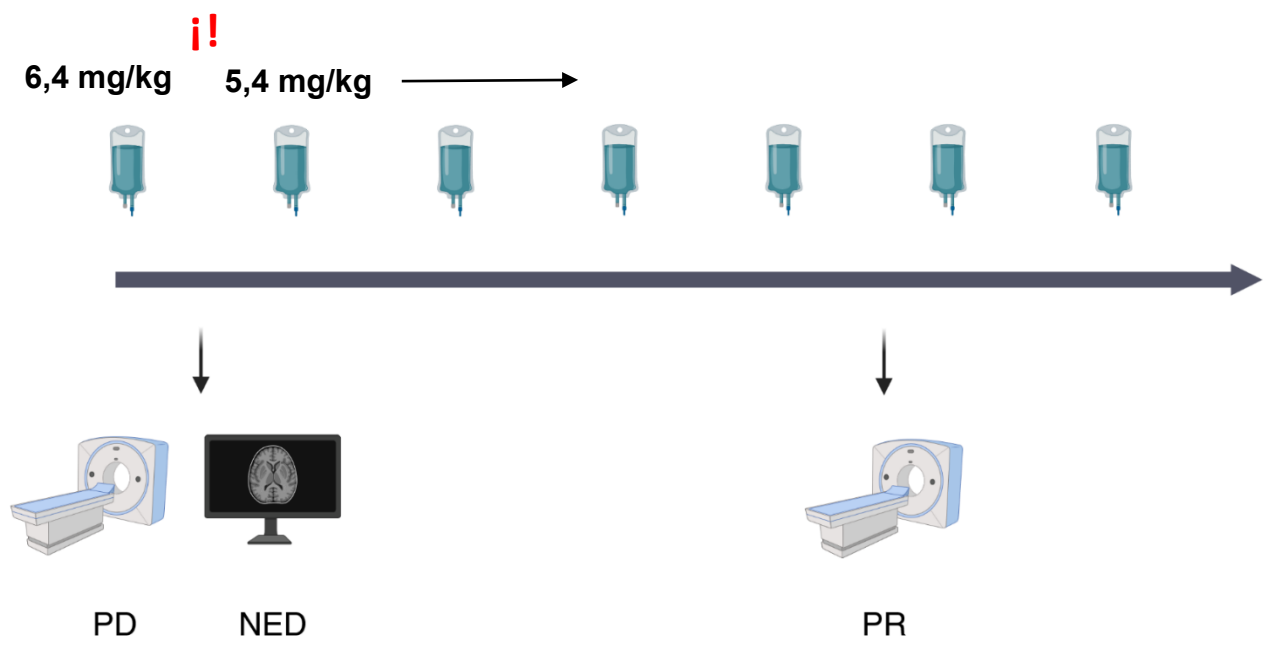
Trastuzumab deruxtecan



CT scan



Brain MRI



Take home messages

- Biomarker testing is key in order to adequately select the treatment for advanced gastric cancer.
- Chemotherapy doublet + trastuzumab (**ToGA**) ± pembrolizumab (in CPS ≥ 1 tumors; **KEYNOTE-811**) is the first line standard of care for HER2-positive advanced gastric cancer patients.
- The continuation of trastuzumab beyond progression is not effective (**T-ACT Study**). Trastuzumab-deruxtecan is the recommended 2nd-line treatment for patients with HER2+ tumors that progress to first line (**DESTINY-Gastric01&02**).
- Brain metastases management should be discussed in a Multidisciplinary Tumor Board. Surgery followed by RT is an option for selected patients. T-Dxd has demonstrated anti-tumor activity against BC brain metastasis.